



*Mechanical
Engineering
Department*

**KONYA
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ENGINEERING MECHANICS STATICS

| SIXTH EDITION |

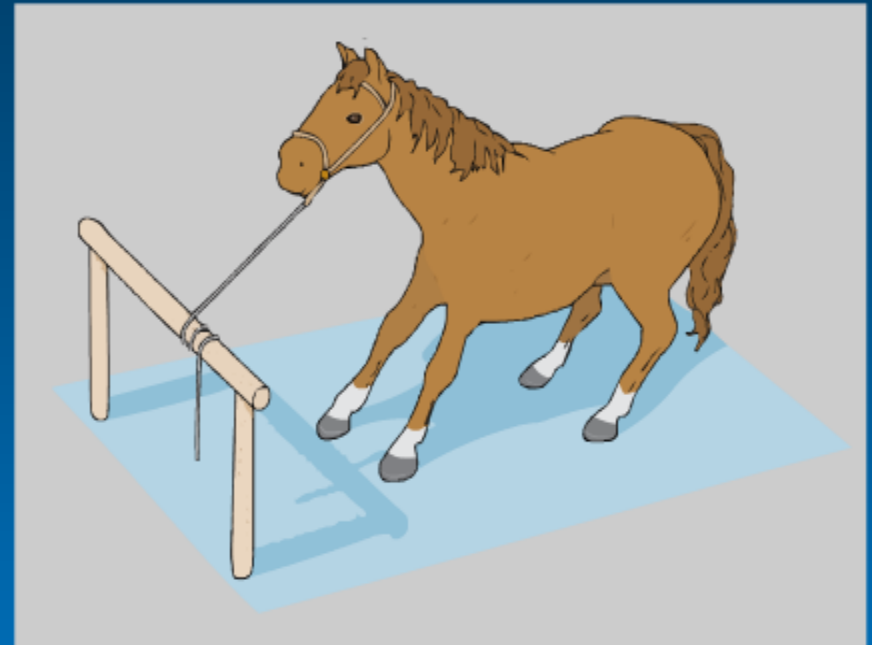
J. L. MERIAM

L. G. KRAIGE

6. Friction

Dry (Coulomb) Friction

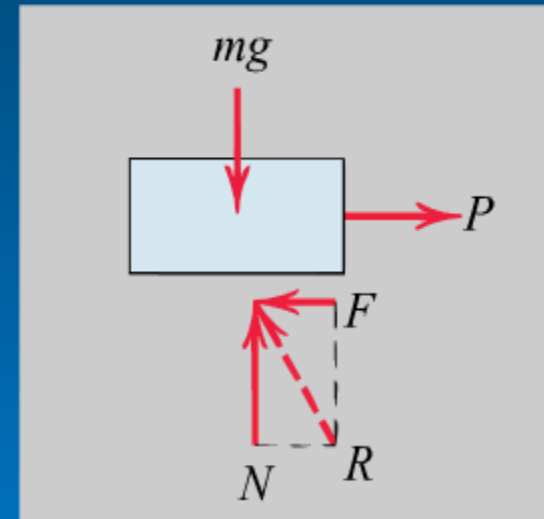
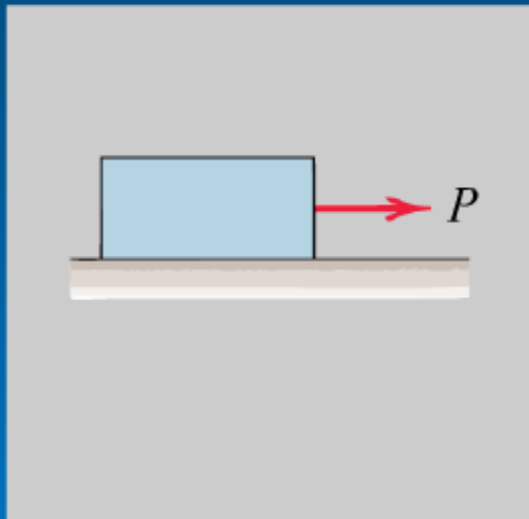
Belt Friction



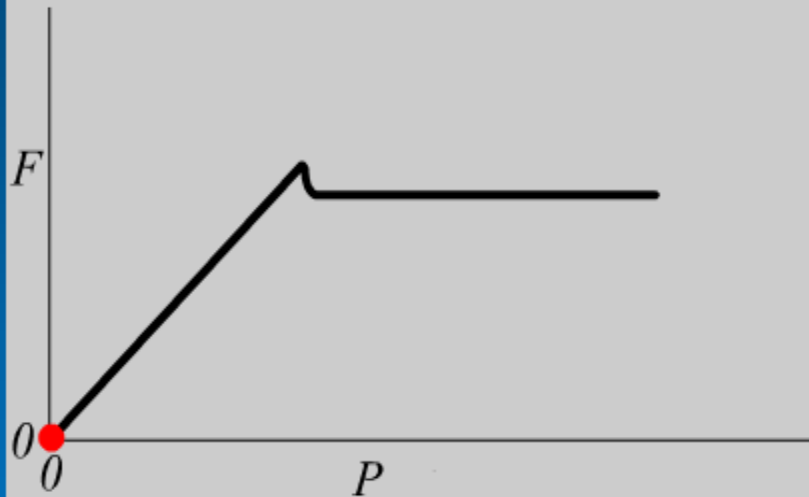
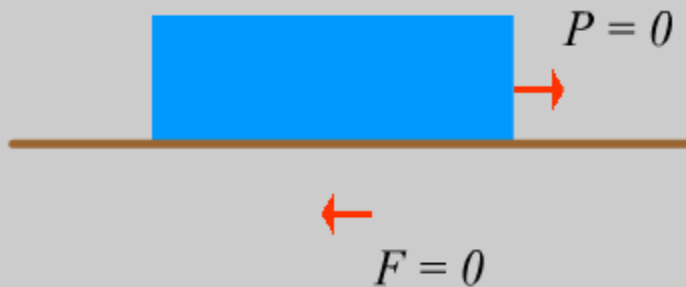
Dry (Coulomb) Friction

Tangential forces called *friction forces* can be generated between all real (rough) surfaces.

Consider a block resting on a rough horizontal surface and let the applied force P increase as indicated in the figure.



$$\Sigma F_x = 0 : P - F = 0 \quad P = F$$

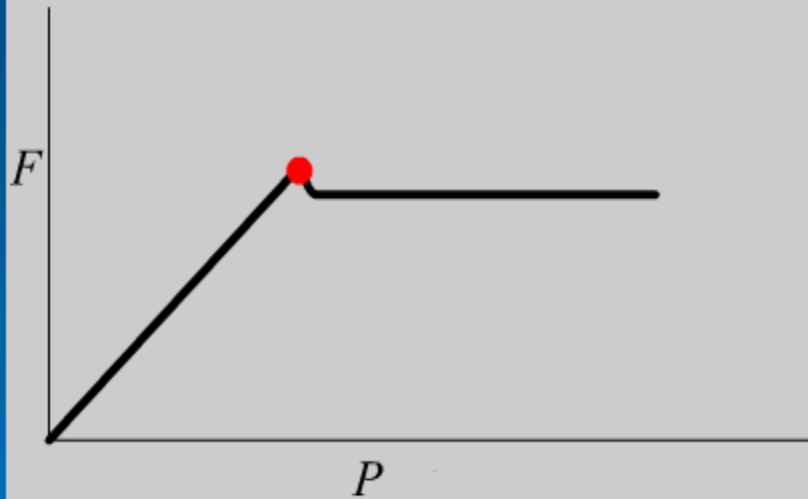
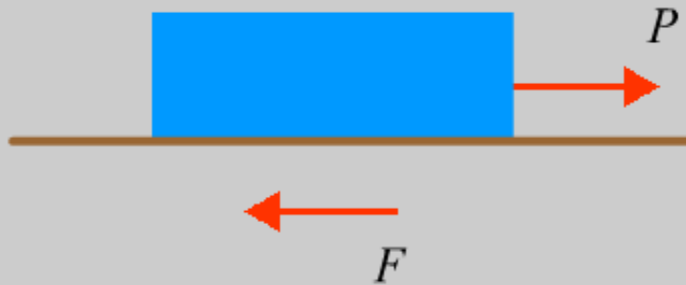


For small values of P :

The value of the resisting friction force F matches P pound for pound, so that the block does not slide. There is no net horizontal force.

Animation

$$\Sigma F_x = 0 : P - F = 0 \quad P = F$$

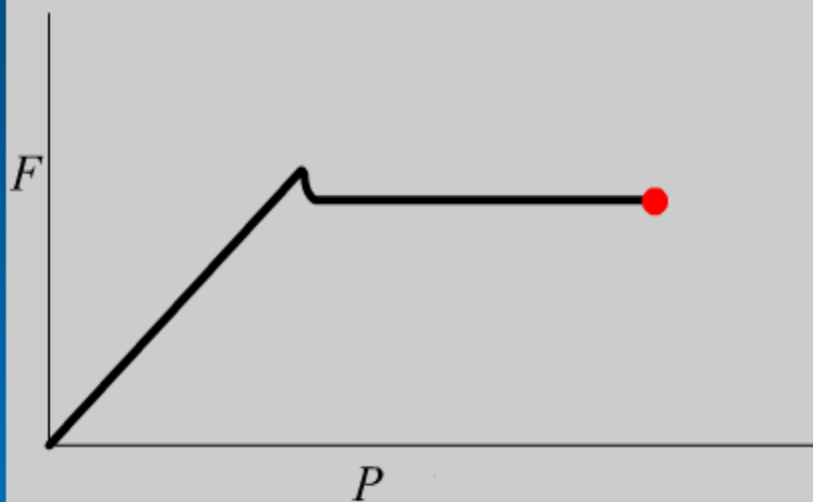
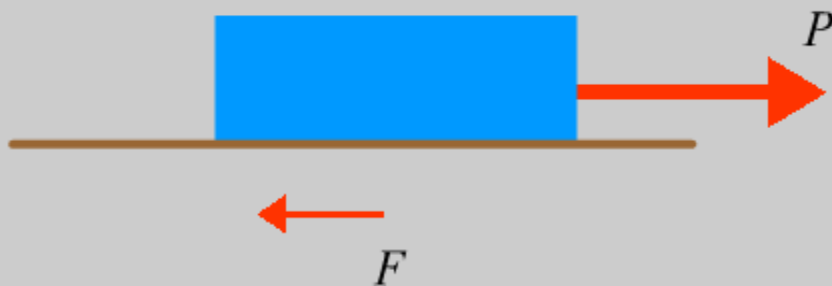


For small values of P :

The value of the resisting friction force F matches P pound for pound, so that the block does not slide. There is no net horizontal force.

Continue

$$\Sigma F_x = ma : P - F = ma \quad P > F$$



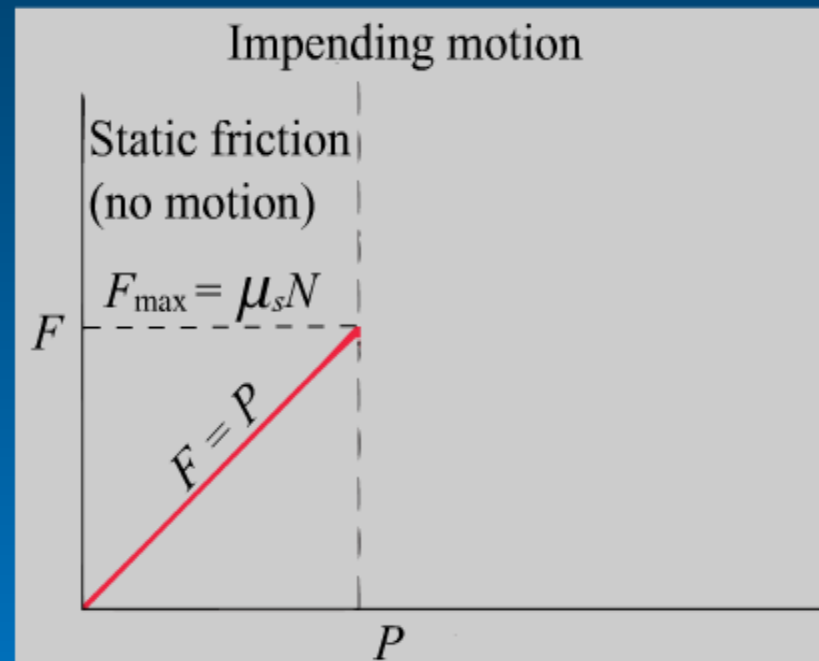
For large values of P :

The maximum value of F has been exceeded, so that friction force "gives up" and the block accelerates to the right due to a net horizontal force.

The maximum value of the friction force at which it "gives up" is called $F_{\max}$ and is given by

$$F_{\max} = \mu_s N$$

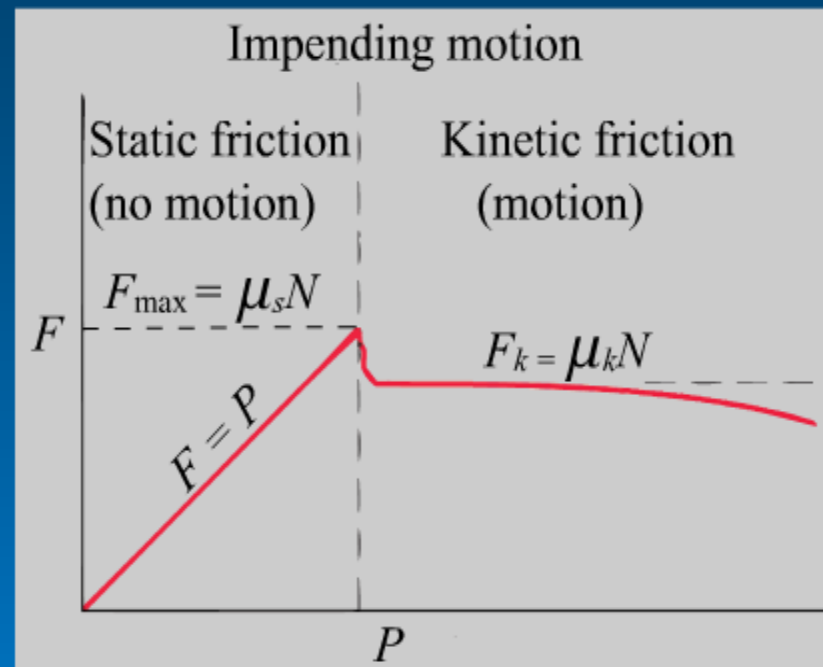
where μ_s is the coefficient of static friction and N is the local normal force.



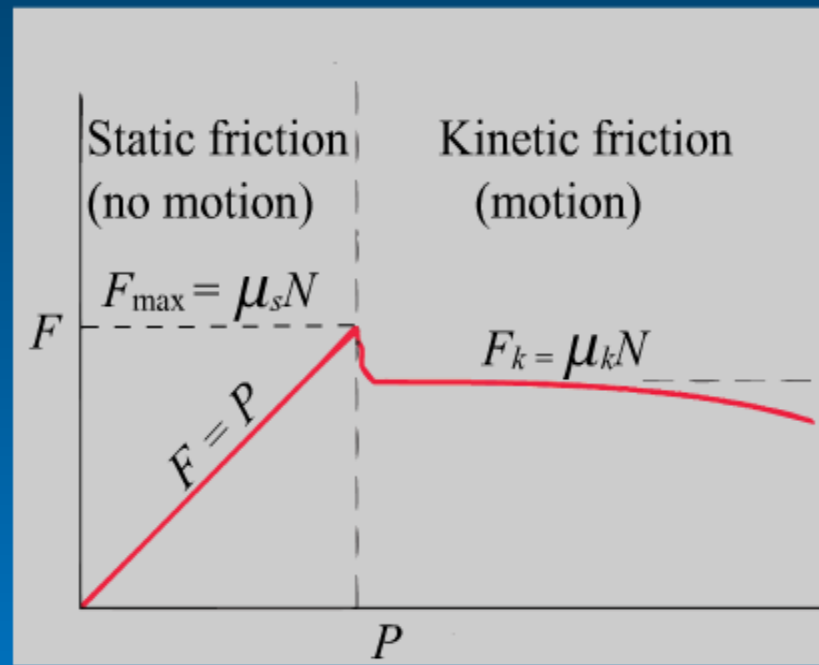
Once motion begins, the friction force is called F_k and is given by

$$F_k = \mu_k N$$

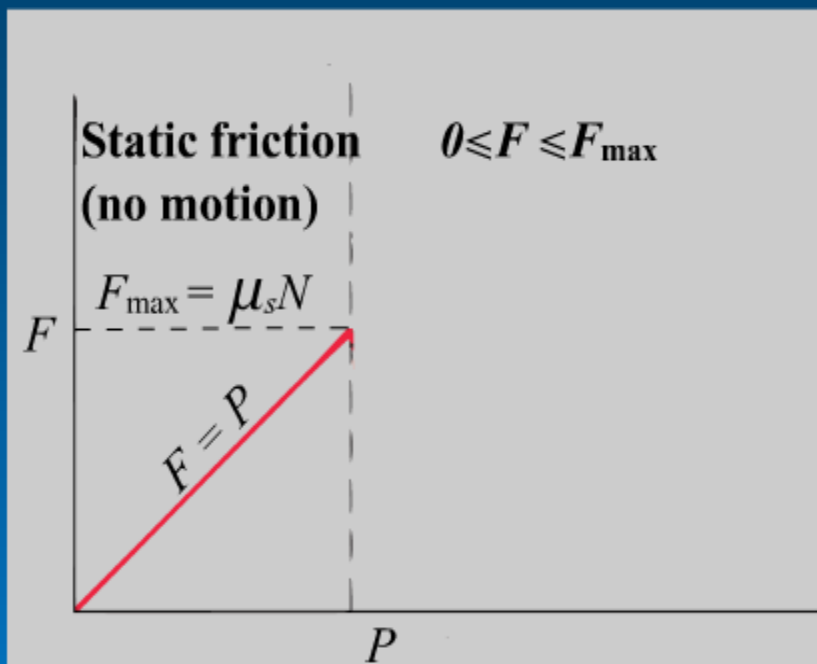
where μ_k is the coefficient of kinetic friction.



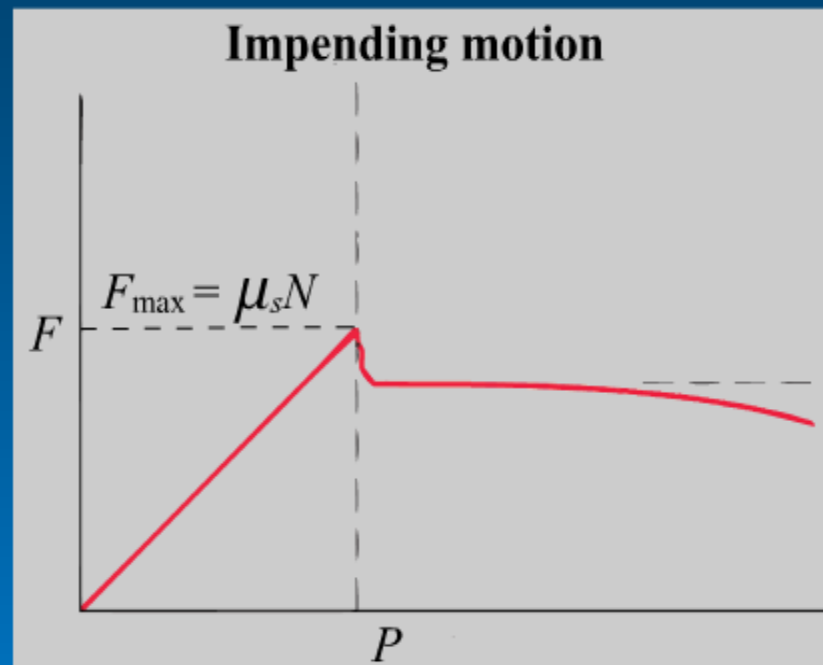
Note that μ_k is usually a bit smaller than μ_s , meaning that, once motion begins, less force P is required to maintain the motion than was required to initiate the motion.



It is critical to understand that the friction force associated with no slipping may take on *any* value between zero and $F_{\max}$.



A frequent error is to assign the value $\mu_s N$ to the friction force F when in fact F may be less than this maximum value. The value of P which causes the maximum static friction force marks the onset of motion, a condition called *impending motion*. Only then do we set $F = F_{\max} = \mu_s N$.



Summary

1. Impending motion is known to exist.

You may set $F = F_{\max} = \mu_s N$ and use equilibrium equations to determine any unknown.

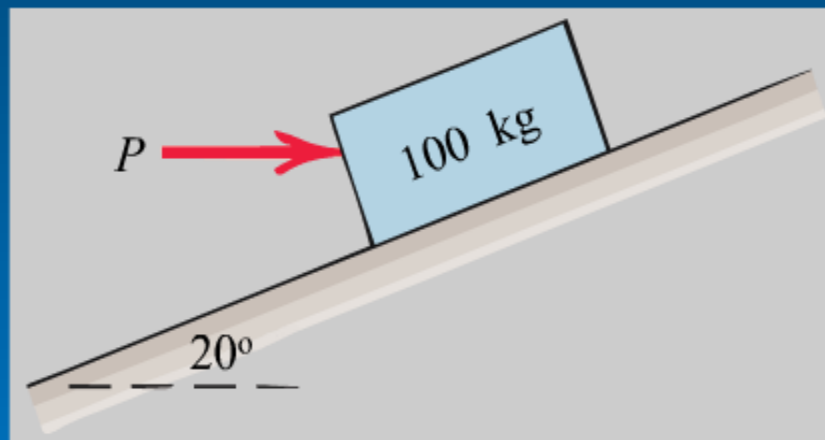
2. We do not know whether motion, impending motion, or neither, exists. So we *assume* equilibrium, call the unknown friction force F (not $F_{\max}$!), and use equilibrium equations to determine F . Then compare F to $F_{\max} = \mu_s N$.
- (a) $F < F_{\max}$: Assumption of equilibrium is OK. The friction force value is F .
- (b) $F = F_{\max}$: Assumption of equilibrium is OK. But motion impends. The value of the friction force is $F_{\max}$.
- (c) $F > F_{\max}$: This cannot be, so our assumption of equilibrium is not valid. The value of the friction force is $F = F_k = \mu_k N$.

3. Slipping is known to occur, so we set $F = F_k = \mu_k N$

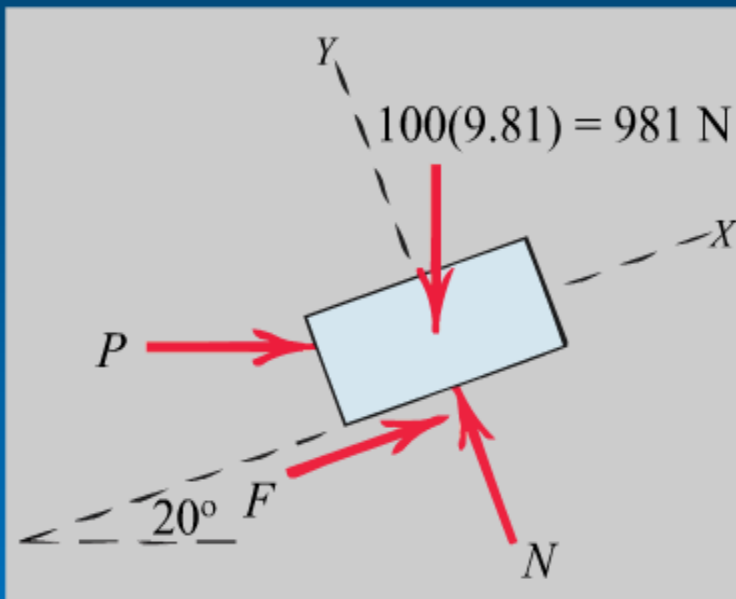
Categories 1 and 3 are easy; the second requires the most care.

Example 1

Determine the magnitude and direction of the friction force acting on the 100-kg block shown if, (a) $P = 500$ N, and (b) $P = 100$ N. The coefficient of static friction is 0.20, and the coefficient of kinetic friction is 0.17. The forces are applied with the block initially at rest.



Because we do not know whether motion impends or occurs, we have a Category 2 problem.



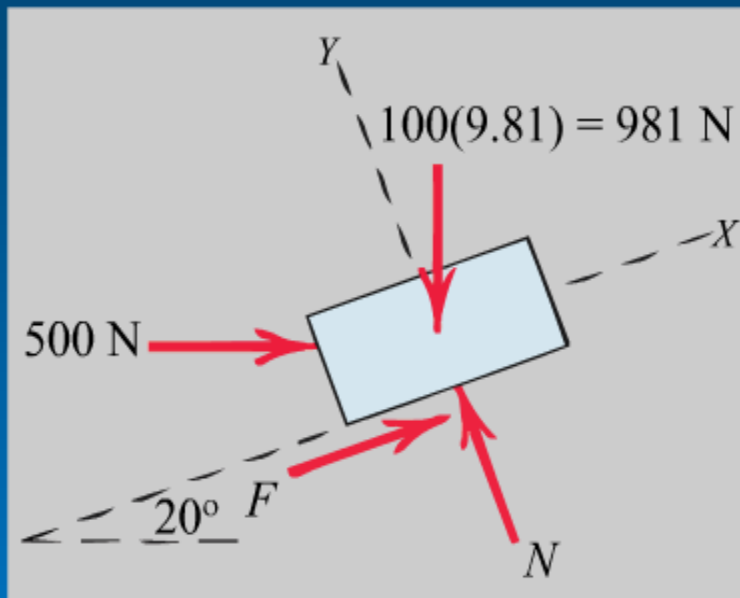
We assume an unknown friction force F acting up the incline and also assume static equilibrium.

(a) $P = 500 \text{ N}$

$$\Sigma F_x = 0 :$$

$$\Sigma F_y = 0 :$$

Solution :

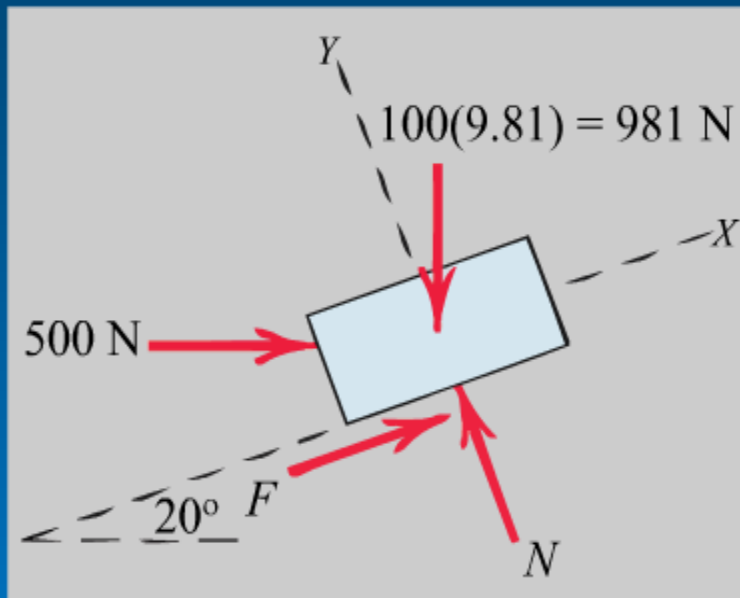


(a) $P = 500 \text{ N}$

$$\Sigma F_x = 0 : 500 \cos 20^\circ + F - 981 \sin 20^\circ = 0$$

$$\Sigma F_y = 0 :$$

Solution :

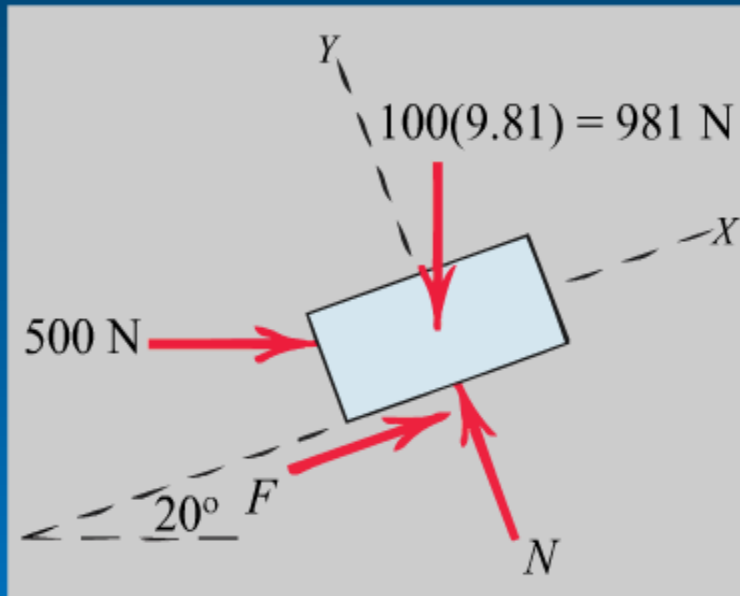


(a) $P = 500 \text{ N}$

$$\Sigma F_x = 0 : 500 \cos 20^\circ + F - 981 \sin 20^\circ = 0$$

$$\Sigma F_y = 0 : N - 500 \sin 20^\circ - 981 \cos 20^\circ = 0$$

Solution :



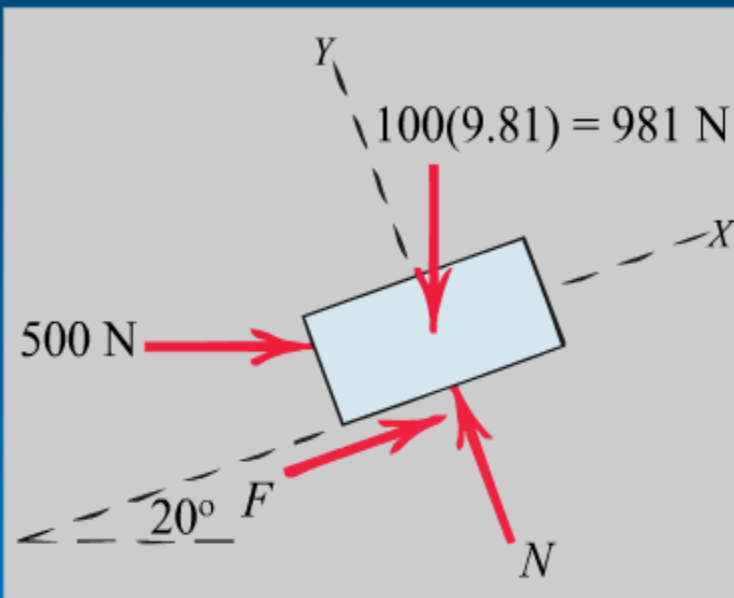
(a) $P = 500 \text{ N}$

$$\Sigma F_x = 0 : 500 \cos 20^\circ + F - 981 \sin 20^\circ = 0$$

$$\Sigma F_y = 0 : N - 500 \sin 20^\circ - 981 \cos 20^\circ = 0$$

Solution : $F = -134.3 \text{ N}$, $N = 1093 \text{ N}$

Discussion



The minus sign on F indicates that 134.3 N of friction force must act down the incline in order to hold the block in static equilibrium.

The major issue: Is 134.3 N of

(a) $P = 500 \text{ N}$

$$\Sigma F_x = 0 : 500 \cos 20^\circ + F - 981 \sin 20^\circ = 0$$

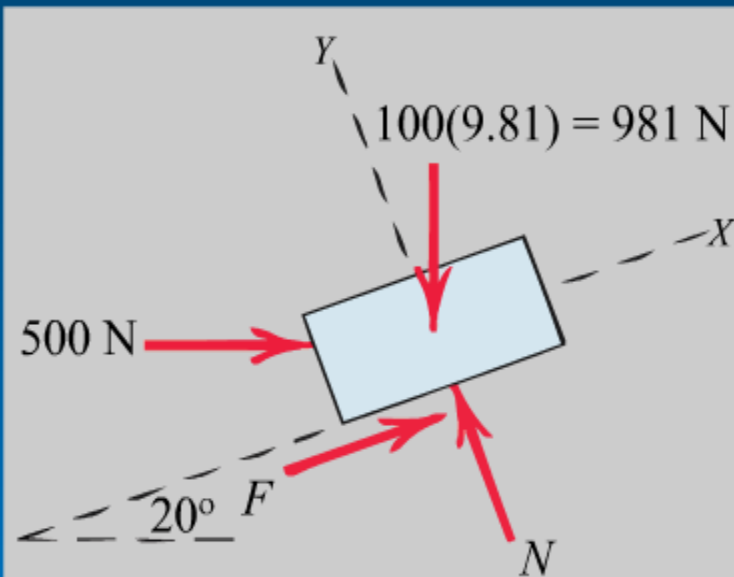
$$\Sigma F_y = 0 : N - 500 \sin 20^\circ - 981 \cos 20^\circ = 0$$

Solution : $F = -134.3 \text{ N}$, $N = 1093 \text{ N}$

Discussion

The major issue: Is 134.3 N of friction force available?

To answer this question,
we compute $F_{\max} = \mu_s N$
 $= 0.20(1093) = 219 \text{ N}$



(a) $P = 500 \text{ N}$

$$\Sigma F_x = 0 : 500 \cos 20^\circ + F - 981 \sin 20^\circ = 0$$

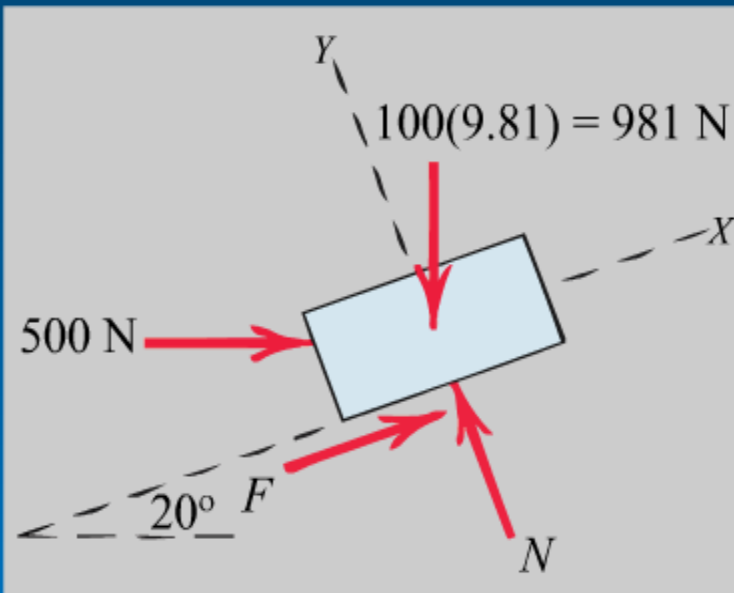
$$\Sigma F_y = 0 : N - 500 \sin 20^\circ - 981 \cos 20^\circ = 0$$

Solution : $F = -134.3 \text{ N}$, $N = 1093 \text{ N}$

Discussion

Since $|F| < F_{\max}$, a 134.3 N friction force is available. Our assumption of static equilibrium is valid, and the answer to the question is

$F = 134.3 \text{ N}$ down the incline.

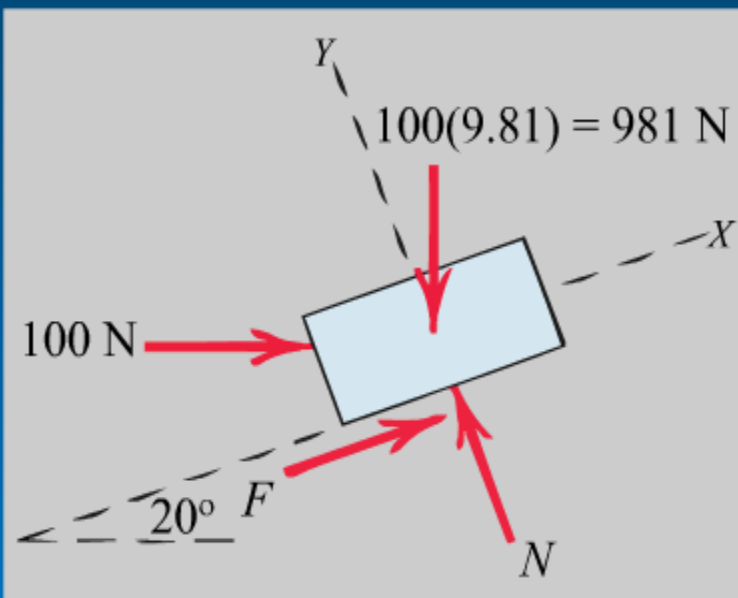


(b) $P = 100 \text{ N}$

$$\Sigma F_x = 0 :$$

$$\Sigma F_y = 0 :$$

Solution :

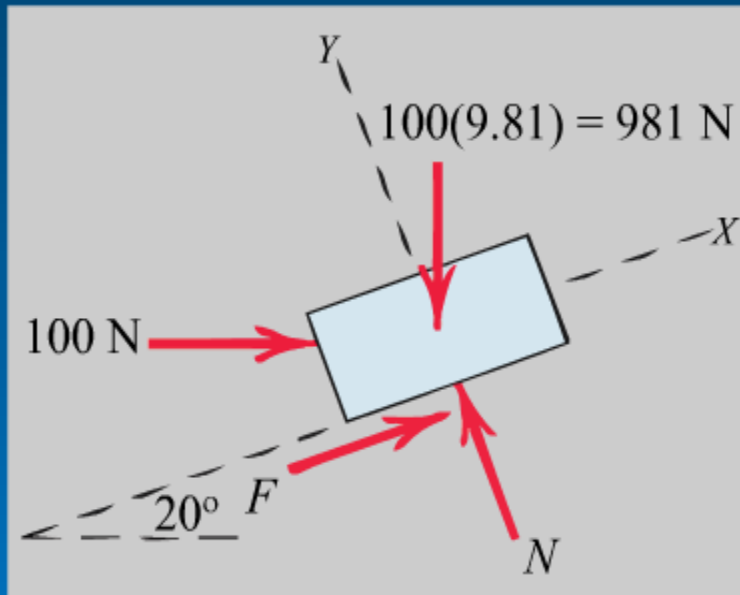


(b) $P = 100 \text{ N}$

$$\Sigma F_x = 0 : 100 \cos 20^\circ + F - 981 \sin 20^\circ = 0$$

$$\Sigma F_y = 0 :$$

Solution :

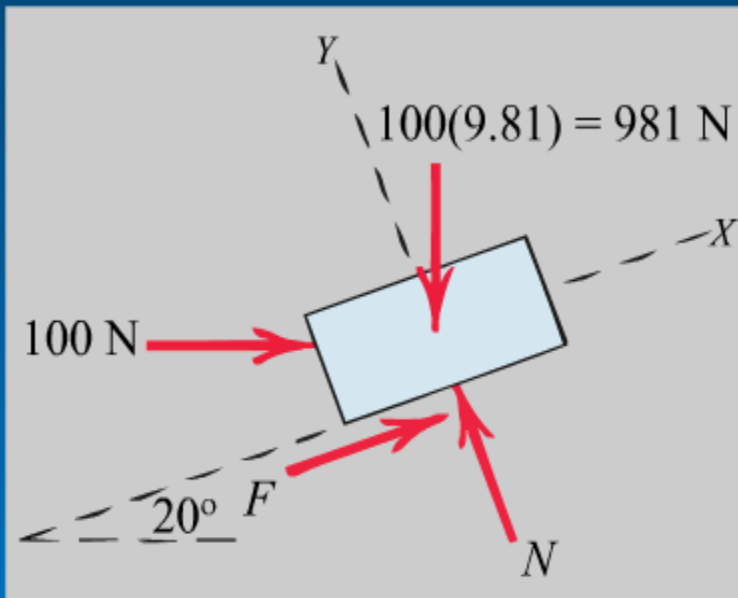


(b) $P = 100 \text{ N}$

$$\Sigma F_x = 0 : 100 \cos 20^\circ + F - 981 \sin 20^\circ = 0$$

$$\Sigma F_y = 0 : N - 100 \sin 20^\circ - 981 \cos 20^\circ = 0$$

Solution :



(b) $P = 100 \text{ N}$

$$\Sigma F_x = 0 : 100 \cos 20^\circ + F - 981 \sin 20^\circ = 0$$

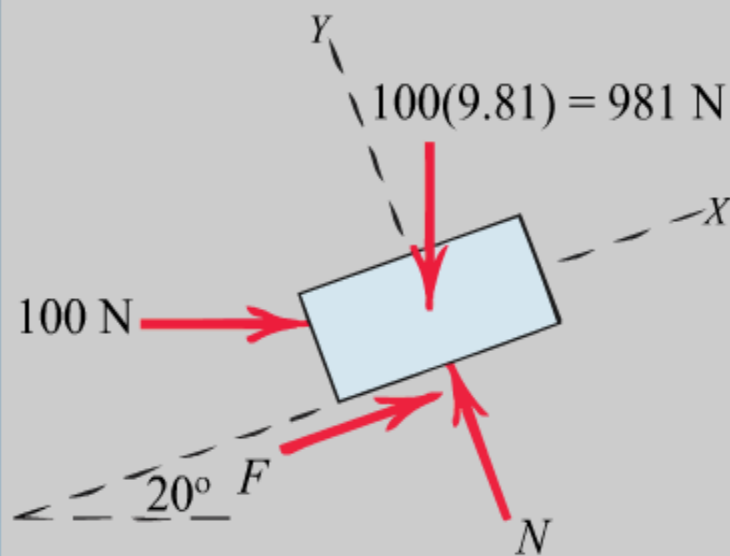
$$\Sigma F_y = 0 : N - 100 \sin 20^\circ - 981 \cos 20^\circ = 0$$

Solution : $F = 242 \text{ N}$, $N = 956 \text{ N}$

Discussion

The positive sign on F indicates that our assumed direction on the FBD is correct.

Again the major issue: Is 242 N of friction force available?



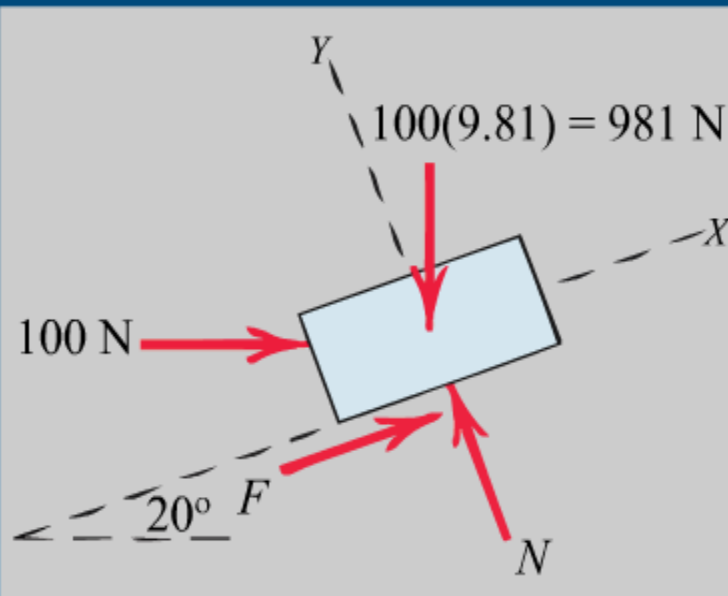
(b) $P = 100 \text{ N}$

$$\Sigma F_x = 0 : 100 \cos 20^\circ + F - 981 \sin 20^\circ = 0$$

$$\Sigma F_y = 0 : N - 100 \sin 20^\circ - 981 \cos 20^\circ = 0$$

Solution : $F = 242 \text{ N}, N = 956 \text{ N}$

Discussion



To answer this question, we compute $F_{\max} = \mu_s N = 0.20(956) = 191.2 \text{ N}$

Since $F > F_{\max}$, a 242-N friction force is NOT available. Our assumption of static equilibrium

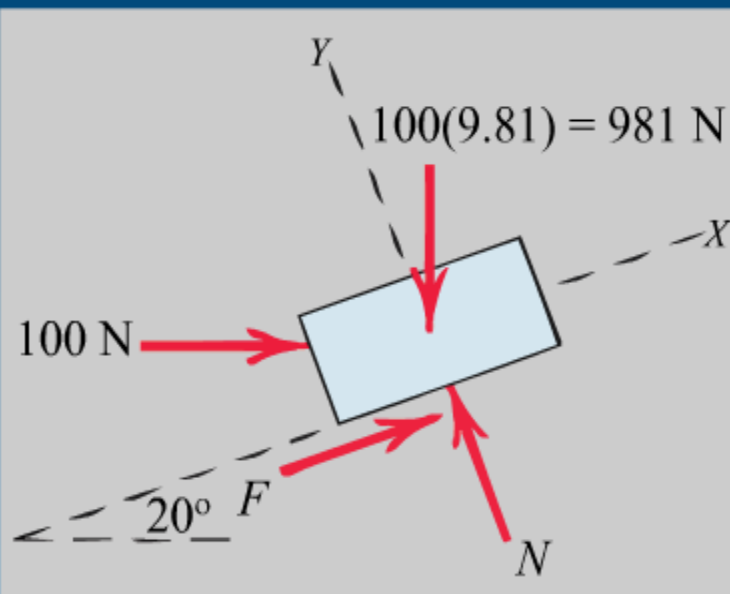
(b) $P = 100 \text{ N}$

$$\Sigma F_x = 0 : 100 \cos 20^\circ + F - 981 \sin 20^\circ = 0$$

$$\Sigma F_y = 0 : N - 100 \sin 20^\circ - 981 \cos 20^\circ = 0$$

Solution : $F = 242 \text{ N}, N = 956 \text{ N}$

Discussion



assumption of static equilibrium is NOT valid, the block slips down the incline and the answer to the question is

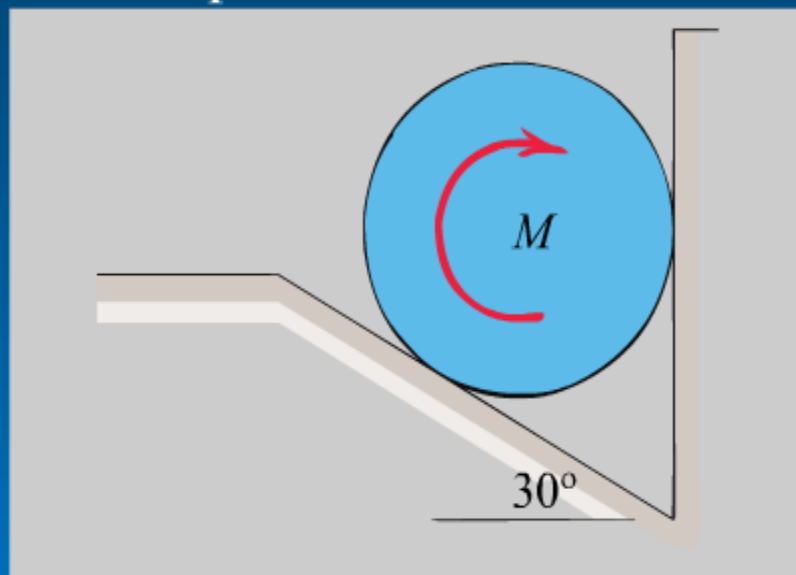
$$F = \mu_k N = 0.17(956) = 162.5 \text{ N}$$

up the incline



Example 2

The 30-kg homogeneous cylinder of 400-mm diameter rests against the surfaces shown. The coefficient of static friction at all interfaces is $\mu_s = 0.25$. Determine the value of M which will cause the cylinder to slip.



$$\Sigma F_x = 0 :$$

$$\Sigma F_y = 0 :$$

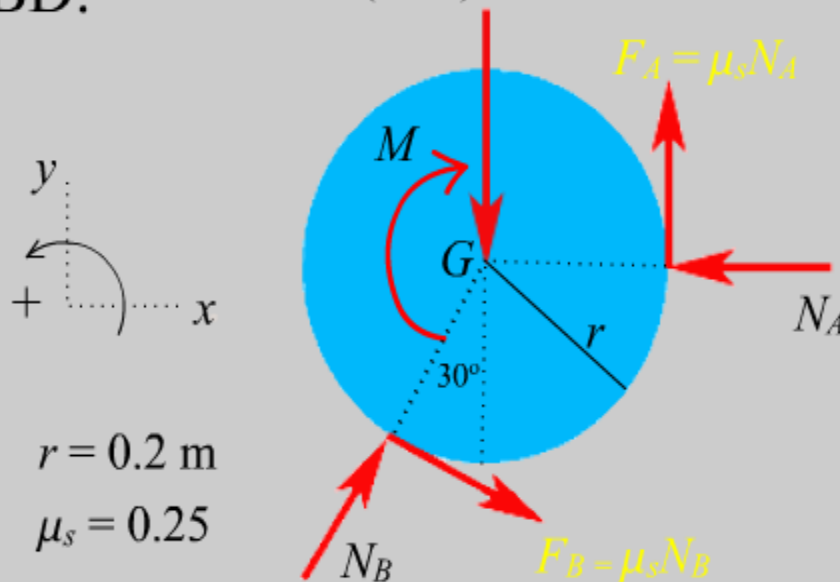
$$\Sigma M_G = 0 :$$

Unknowns:

FBD:

$$30(9.81) = 294 \text{ N}$$

Solution:



$$r = 0.2 \text{ m}$$

$$\mu_s = 0.25$$

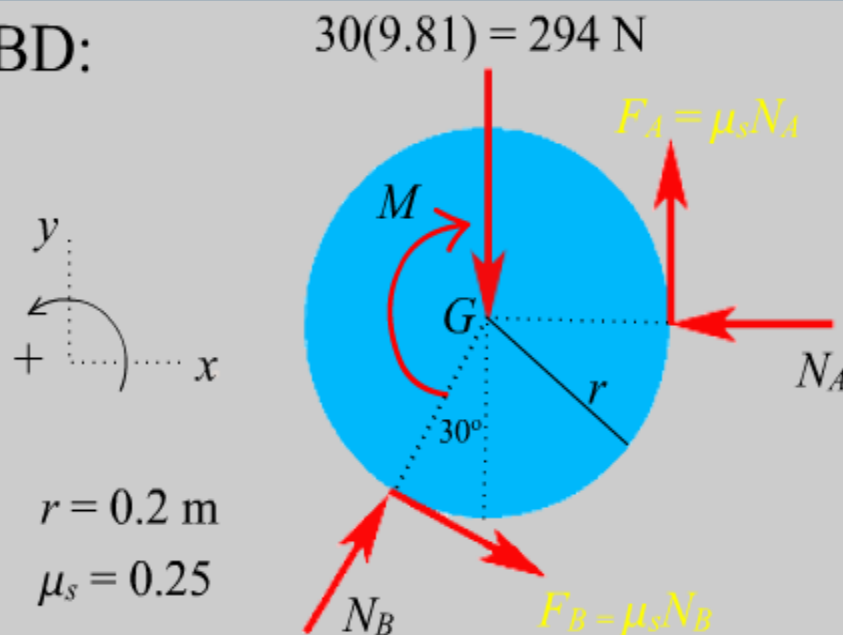
$$\Sigma F_x = 0 : N_B \sin 30^\circ + 0.25 N_B \cos 30^\circ - N_A = 0$$

$$\Sigma F_y = 0 :$$

$$\Sigma M_G = 0 :$$

Unknowns:

FBD:



Solution:

$$\Sigma F_x = 0 : N_B \sin 30^\circ + 0.25 N_B \cos 30^\circ - N_A = 0$$

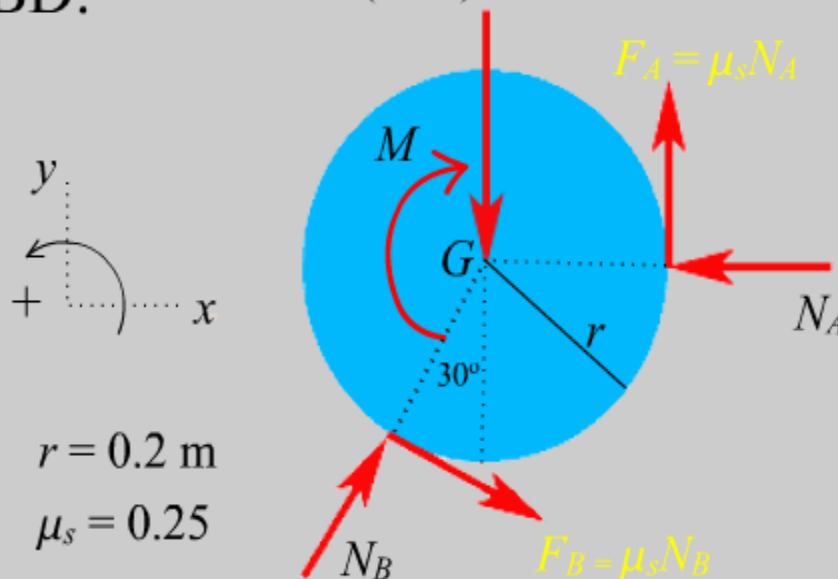
$$\Sigma F_y = 0 : N_B \cos 30^\circ - 0.25 N_B \sin 30^\circ + 0.25 N_A - 294 = 0$$

$$\Sigma M_G = 0 :$$

Unknowns:

FBD:

$$30(9.81) = 294 \text{ N}$$



Solution:

$$r = 0.2 \text{ m}$$

$$\mu_s = 0.25$$



17/17



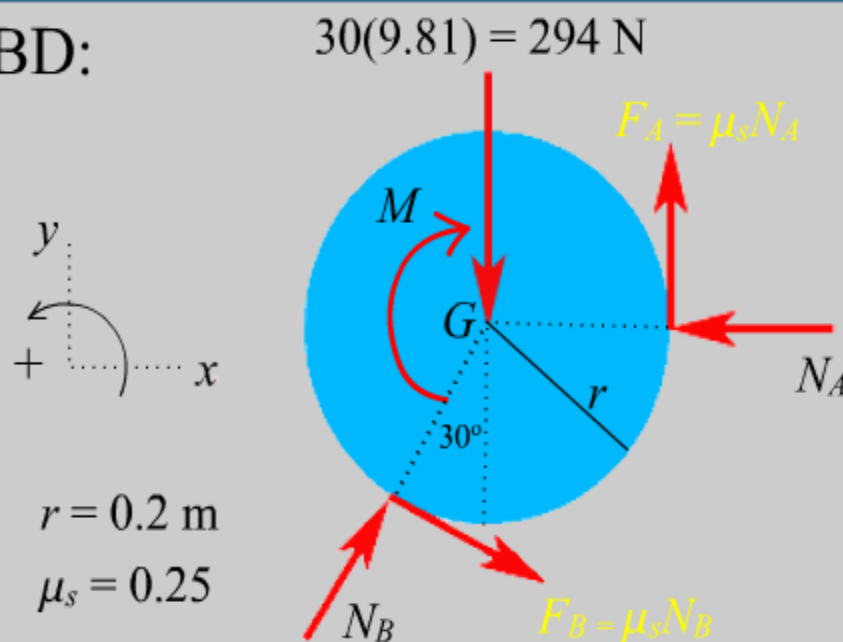
$$\Sigma F_x = 0 : N_B \sin 30^\circ + 0.25 N_B \cos 30^\circ - N_A = 0$$

$$\Sigma F_y = 0 : N_B \cos 30^\circ - 0.25 N_B \sin 30^\circ + 0.25 N_A - 294 = 0$$

$$\Sigma M_G = 0 : -M + 0.25 N_B (0.2) + 0.25 N_A (0.2) = 0$$

Unknowns:

FBD:



Solution:



$$\Sigma F_x = 0 : N_B \sin 30^\circ + 0.25 N_B \cos 30^\circ - N_A = 0$$

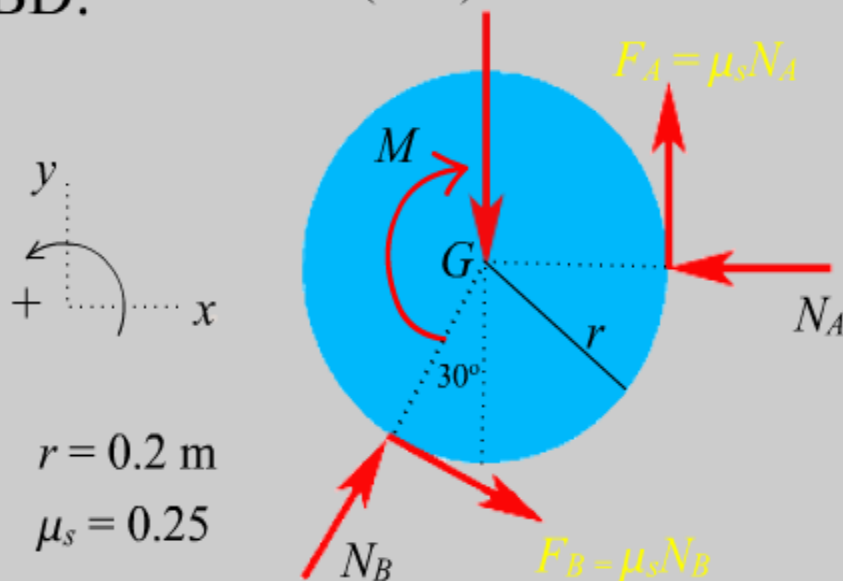
$$\Sigma F_y = 0 : N_B \cos 30^\circ - 0.25 N_B \sin 30^\circ + 0.25 N_A - 294 = 0$$

$$\Sigma M_G = 0 : -M + 0.25 N_B (0.2) + 0.25 N_A (0.2) = 0$$

Unknowns: N_A, N_B, M

FBD:

$$30(9.81) = 294 \text{ N}$$



Solution:

$$r = 0.2 \text{ m}$$

$$\mu_s = 0.25$$

$$\Sigma F_x = 0 : N_B \sin 30^\circ + 0.25 N_B \cos 30^\circ - N_A = 0$$

$$\Sigma F_y = 0 : N_B \cos 30^\circ - 0.25 N_B \sin 30^\circ + 0.25 N_A - 294 = 0$$

$$\Sigma M_G = 0 : -M + 0.25 N_B (0.2) + 0.25 N_A (0.2) = 0$$

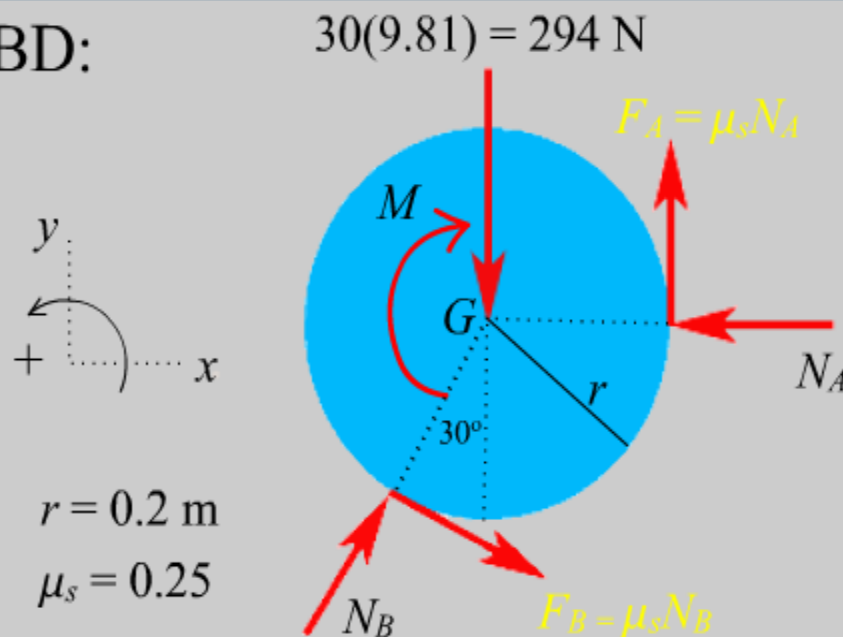
Unknowns: N_A, N_B, M

$$N_A = 191.6 \text{ N}$$

Solution: $N_B = 268 \text{ N}$

$$M = 22.9 \text{ N}\cdot\text{m}$$

FBD:



$$\Sigma F_x = 0 : N_B \sin 30^\circ + 0.25 N_B \cos 30^\circ - N_A = 0$$

$$\Sigma F_y = 0 : N_B \cos 30^\circ - 0.25 N_B \sin 30^\circ + 0.25 N_A - 294 = 0$$

$$\Sigma M_G = 0 : -M + 0.25 N_B (0.2) + 0.25 N_A (0.2) = 0$$

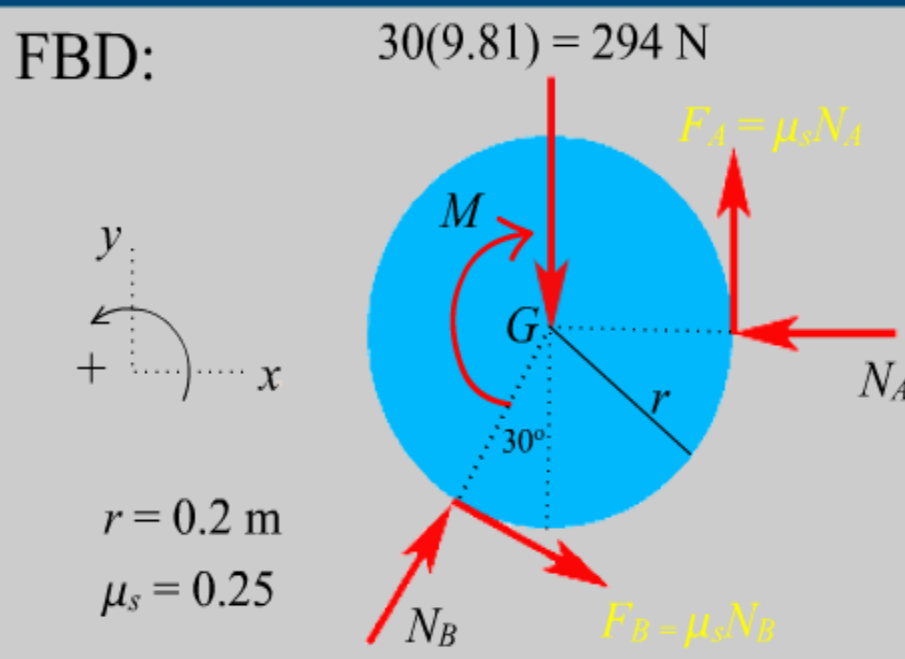
Unknowns: N_A, N_B, M

$$N_A = 191.6 \text{ N}$$

Solution: $N_B = 268 \text{ N}$

$$M = 22.9 \text{ N}\cdot\text{m}$$

We may set the friction forces to their maximum possible values because the problem involves *impending motion*.

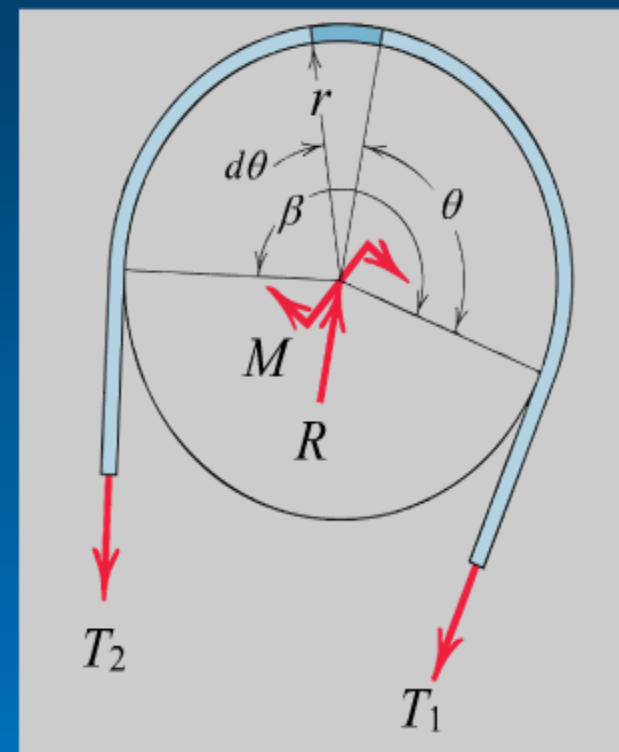


Belt Friction

Consider a flexible belt, rope, or cable wrapped around a fixed drum as shown in the figure.

If there is friction between the belt and the fixed drum, then the tension T_2 at one end of the belt can differ from that (T_1) at the other end.

We consider a FBD of an element of the belt. We assume that slipping impends in the direction of T_2 .



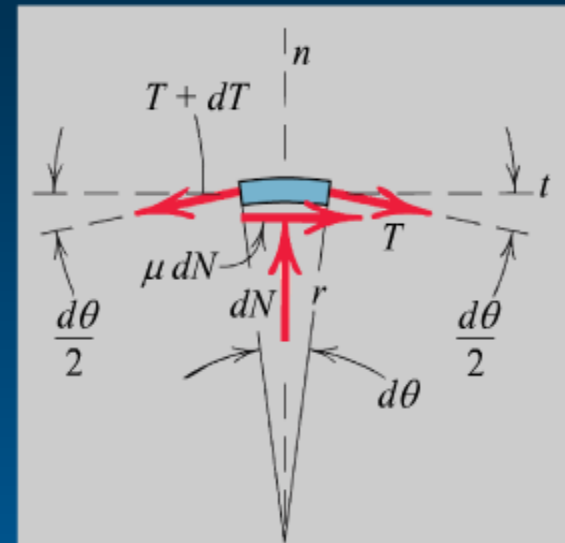
$$\Sigma F_t = 0 :$$

$$T \cos \frac{d\theta}{2} + \mu dN - (T+dT) \cos \frac{d\theta}{2} = 0$$

$$\text{or } \mu dN = dT \dots\dots\dots (1)$$

where we have assumed small angles
so that

$$\cos \frac{d\theta}{2} \cong 1$$



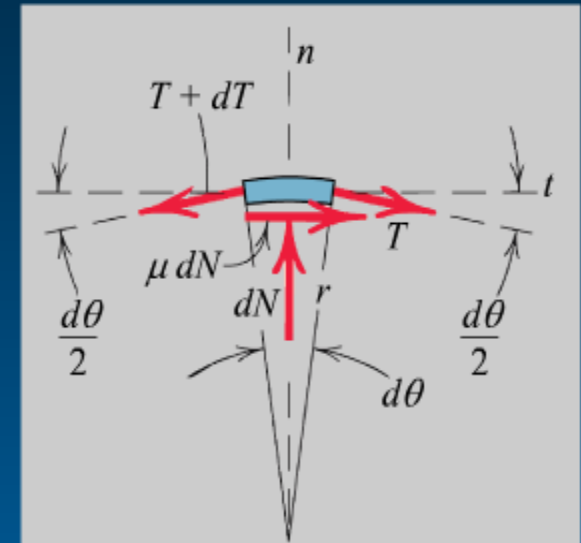
$$\Sigma F_n = 0 :$$

$$dN - T \sin \frac{d\theta}{2} - (T+dT) \sin \frac{d\theta}{2} = 0$$

$$\text{or } dN = T d\theta \dots\dots\dots (2)$$

where we have ignored the product of two differentials compared to first-order terms and set

$$\sin \frac{d\theta}{2} \cong \frac{d\theta}{2} \quad \text{for small angles}$$



$$\mu dN = dT \dots\dots\dots (1) \qquad dN = T d\theta \dots\dots\dots (2)$$

Considering Eqs. (1) & (2) gives

$$\frac{dT}{T} = \mu d\theta$$

Integrating both sides

$$\int_{T_1}^{T_2} \frac{dT}{T} = \int_0^{\beta} \mu d\theta \quad \text{or} \quad \frac{T_2}{T_1} = e^{\mu\beta}$$

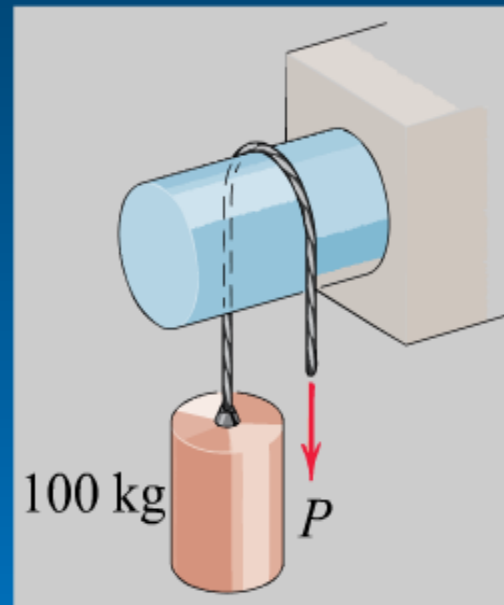
Important Notes: T_2 is the larger tension.

β is the total angle of belt contact *in radians*.



Example

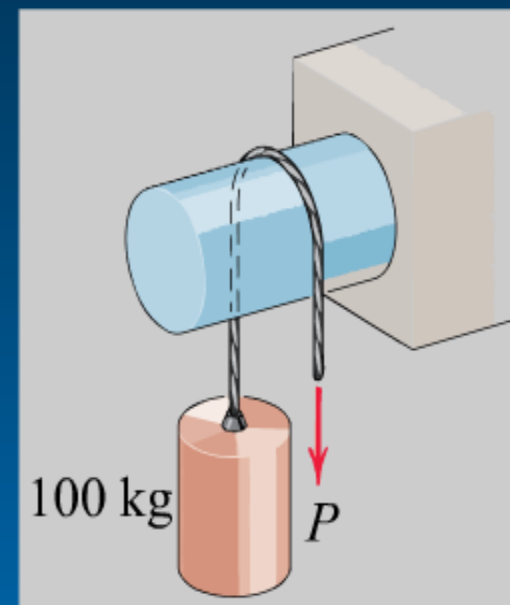
Determine the force P required to (a) raise and (b) lower the 100-kg cylinder at a slow steady speed. The coefficient of friction between the cord and the supporting surface is 0.30.



(a) Motion of cylinder impends up:

$$\frac{T_2}{T_1} = e^{\mu\beta} : \quad \frac{P_{up}}{100(9.81)} = e^{0.3\pi}$$

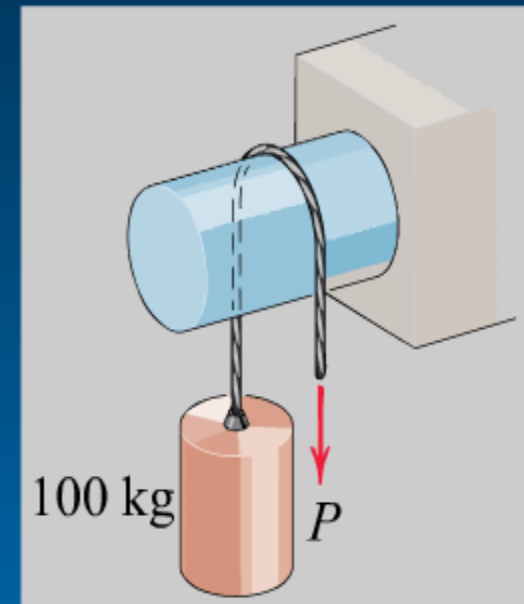
$$P_{up} = 2520 \text{ N}$$



(b) Motion of cylinder impends down:

$$\frac{T_2}{T_1} = e^{\mu\beta} : \quad \frac{100(9.81)}{P_{down}} = e^{0.3\pi}$$

$$P_{down} = 382.3 \text{ N}$$





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